

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Architect GPU/HPC infrastructure for AI workloads: CUDA-based compute clusters with InfiniBand interconnects, orchestrated via Slurm, supporting distributed training and inference at scale.
- Built RAG pipelines with vector search and embedding-based retrieval (Pinecone, FAISS), plus GraphRAG improving LLM response grounding over standard vector RAG for complex, multi-hop queries.
- Designed multi-step LLM evaluation tasks and an experimentation pipeline analyzing model performance and regressions across versions, identifying where and why model behavior degrades before release.
- Hardened the system from working to reliable: built an AI evaluation, monitoring, and observability pipeline enforcing automated compliance, quality, and drift checks, blocking 15% of releases that failed validation.

QureAI (Healthcare AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Extended and optimized inference framework internals (vLLM-class serving systems) with kernel-level development (CUDA/Triton), distributed KV cache management, and speculative decoding to improve LLM serving throughput.
- Led migration of PyTorch-backed model serving to async vLLM with autoscaling, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and Java microservices processing regulated financial transactions for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS), with object-oriented system design.
- Drove a redesign of nightly financial data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning, in a domain requiring auditable, traceable transaction records.
- Led standardization of failure-handling contracts across 25+ services in partnership with cross-functional teams through code reviews and design write-ups, reducing production escalations 60% in one quarter.

Projects

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL, built to keep agentic systems trustworthy at volume.

Distributed KV Store | github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus key-value store in-house with leader election, log replication, and log compaction for fault-tolerant distributed coordination.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed, traceable root cause reports.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, C++, Java, SQL

Inference & Serving: vLLM, Kernel Development (CUDA, Triton), Distributed KV Cache, Speculative Decoding, PyTorch, Inference Framework Optimization

ML/AI Engineering: Reinforcement Learning, Model Evaluation & Experimentation, RAG, Agentic AI, LangGraph, MCP

Infra & Cloud: GPU/HPC Clusters, InfiniBand, Slurm, AWS, Azure, GCP, Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer